Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/18/2022
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Glenn Research Center		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: Changed end date to 12/31/2024, per PI (Ed., 11/28/22) NOTE: Changed end date to 12/31/2022, per PI (Ed., 10/3/19) NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David M.S. (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA-STD-6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 pounds per square inch absolute (psia). Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	We completed a series of concurrent rod blowoff tests at variable gravity levels in the centrifuge drop vehicle in the Zero Gravity Research Facility. Testing was conducted with the 0.318 cm radius polymethyl methacrylate (PMMA) rods in the 8.2 pounds per square inch absolute (psia) lunar exploration vehicle pressure. The flame stagnation region does not seem to be strongly affected by Coriolis forces. The results to date of the experiment compare very well with the blowoff branch of the Burning and Suppression of Solids - II (BASS-II) flammability boundary we published in Combustion and Flame in 2017 (Olson S, Ferkul P. "Microgravity flammability boundary for PMMA rods in axial stagnation flow: Experimental results and energy balance analyses." Combustion and Flame. 2017 Jun;180:217-29). The buoyant flow is estimated using Ub = 25 (g/gEarth)^1/2 so the 1g (Earth g) buoyant flow is 25 cm/s for the stagnation geometry. The boundary is just slightly shifted down from the BASS-II boundary. The close alignment of the two boundaries indicates that the square root scaling appears to be valid. The stable flame at lunar g level and 13.8% O2 is very close to the minimum in the BASS-II flammability boundary. This would seem to support prior research which found materials are more flammable on the Moon than on Mars or the Earth. The results were presented at the American Society for Gravitational and Space Research (ASGSR) and a paper of the results is planned. The Solid Fuel Ignition and Extinction (SoFIE) flight flow sensor was found to be unreliable due to damage it apparently sustained during high pressure Fuel Oxidizer Management Assembly (FOMA) calibrations. A ground flow sensor was fast-tracked to be flight qualified and calibrated. The Combustion Tunnel drop rig was used to calibrate the flow sensor against a known forced flow velocity in normal gravity. A preliminary analysis of the data has been conducted with additional analysis underway after receiving updated calibration data for the critical flow orifices used in the testing.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Olson SL, Neumann ES, Valletta VM, Ogorzaly AL. "Partial gravity flammability boundary for cast PMMA rods in axial stagnation flow." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022